

LAB EXPERIMENTS

Q12) a. Write a C program to Encrypt the message “meet me at the usual place at ten rather than eight

oclock” using the Hill cipher with the key.

[9 4

5 7]

a. Show your calculations and the result.

b. Show the calculations for the corresponding decryption of the ciphertext to recover the original

plaintext.

Aim:

To write a C program to implement the **Hill Cipher** using the key matrix

$$K = \begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$$

Encrypt the message "**meet me at the usual place at ten rather than eight oclock**"

and decrypt the ciphertext to recover the original plaintext.

Algorithm:

1. Start the program.
2. Read the plaintext from the user.
3. Remove spaces and convert all letters to lowercase.
4. If the plaintext length is odd, append 'x'.
5. Convert letters into numbers (a = 0, b = 1, ..., z = 25).
6. Divide the plaintext into pairs of letters.
7. Multiply each pair by the key matrix

$$K = \begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$$

using modulo 26.

8. Convert the resulting numbers back into letters to obtain the ciphertext.

9. For decryption, compute the inverse of the key matrix modulo 26.
10. Multiply each ciphertext pair by the inverse matrix to recover the plaintext.
11. Display the ciphertext and decrypted plaintext.
12. Stop the program.

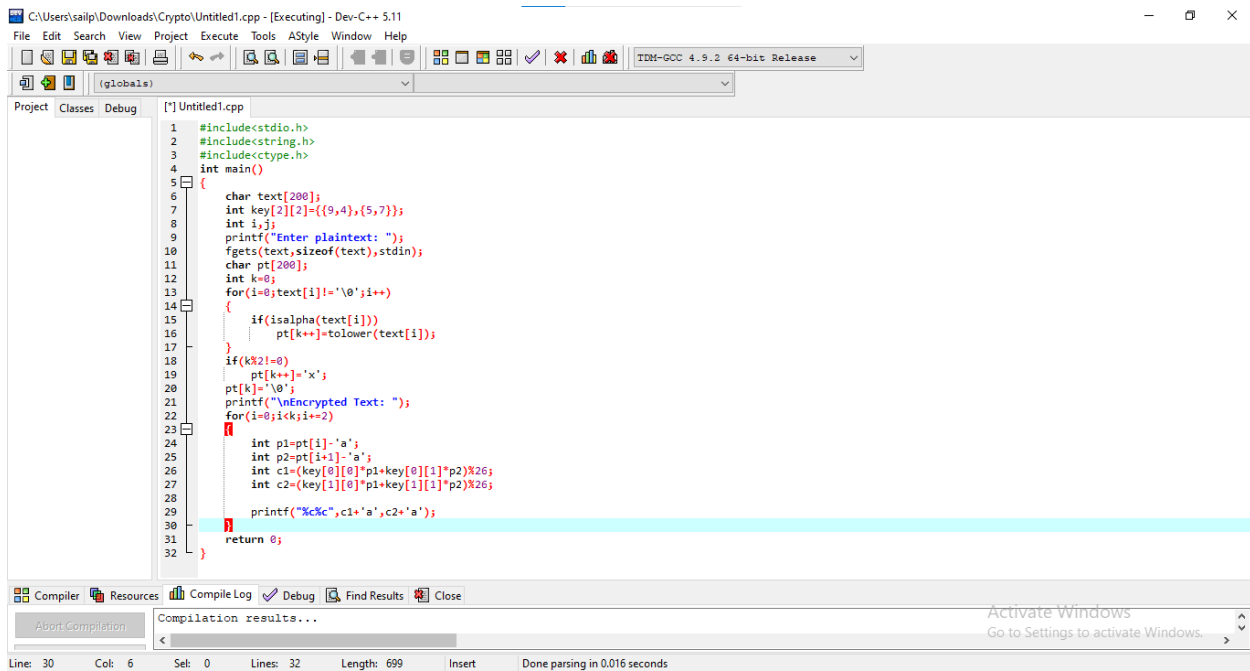
Result:

Thus, the C program to implement the Hill Cipher using the key matrix

$$K = \begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$$

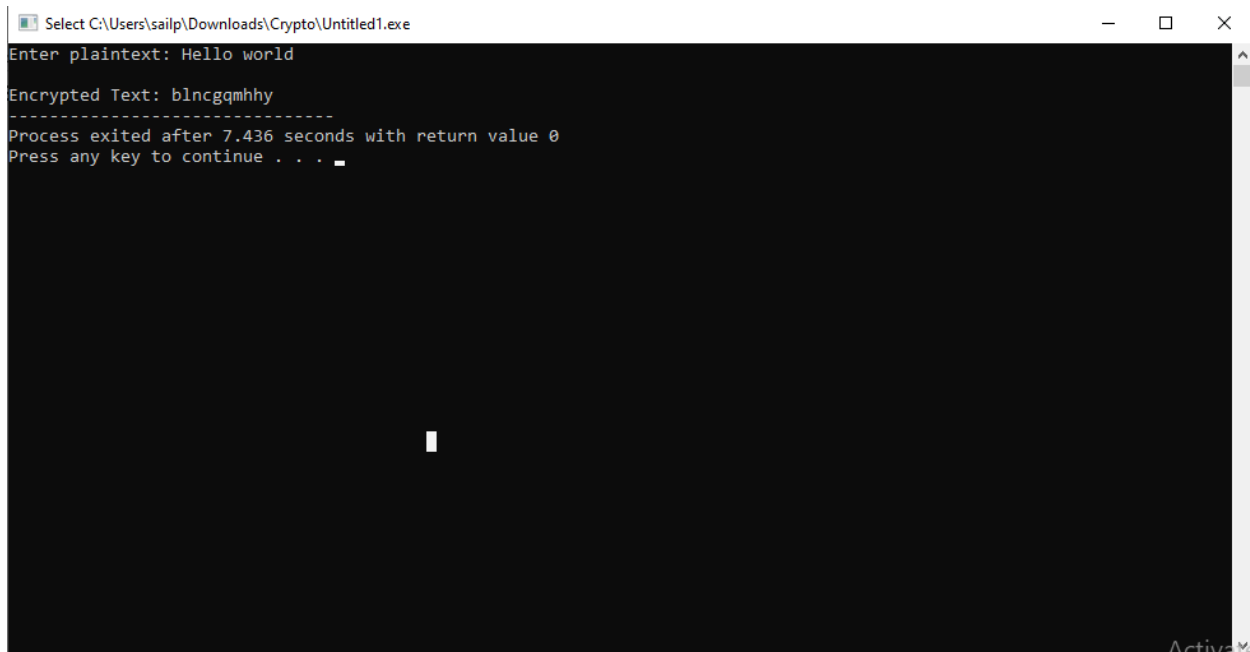
was successfully executed. The plaintext was encrypted into ciphertext using matrix multiplication modulo 26, and the original plaintext was successfully recovered using the inverse key matrix during decryption.

Code:



```
1 #include<stdio.h>
2 #include<string.h>
3 #include<ctype.h>
4 int main()
5 {
6     char text[200];
7     int key[2][2]={{9,4},{5,7}};
8     int i,j;
9     printf("Enter plaintext: ");
10    fgets(text,sizeof(text),stdin);
11    char pt[200];
12    int k=0;
13    for(i=0;text[i]!='\0';i++)
14    {
15        if(isalpha(text[i]))
16            pt[k++]+=tolower(text[i]);
17    }
18    if(k%2!=0)
19        pt[k++]='x';
20    pt[k]='\0';
21    printf("\nEncrypted Text: ");
22    for(i=0;i<k;i+=2)
23    {
24        int p1=pt[i]-'a';
25        int p2=pt[i+1]-'a';
26        int c1=(key[0][0]*p1+key[0][1]*p2)%26;
27        int c2=(key[1][0]*p1+key[1][1]*p2)%26;
28
29        printf("%c%c",c1+'a',c2+'a');
30    }
31    return 0;
32 }
```

Output:



```
Select C:\Users\sailp\Downloads\Crypto\Untitled1.exe
Enter plaintext: Hello world
Encrypted Text: blncgqmhhY
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Process exited after 7.436 seconds with return value 0
Press any key to continue . . .
```